A068 — PKLSA Intrinsic Phase Clock Framework v1.0.8 Scientific Falsifier Protocol

SST-Workbench

Version v0.2.0 — pre-freeze method qualification

Abstract

This protocol incorporates the two user-supplied A068 preflight reports and the immutable A064 v0.3.0 upstream audit into an SST Falsifier Framework v1.0.8 instance. It validates phase-detection numerics and source integrity, *not* a physically generated PKLSA clock. Historical v0.1.0 and the frozen v1.0.7 framework remain untouched. The v1.0.8 runtime owns unique run IDs, blind preservation, report rendering, and fail-closed reveal policy.

1 Identification and status

Catalog: A068; instance: A068-v0.2.0; runtime: SST Falsifier Framework v1.0.8 (local release candidate); profile: `multilibrary_gpu`. Status: DEVELOPMENT / NOT CERTIFIED. Earlier A068 v0.1.0 is an unblinded method preflight. All physical clock claims remain unresolved.

2 Research question and objective

Can a source-native, materially bounded three-dimensional PKLSA vortex dynamically produce a reproducible intrinsic phase frequency, without using an imposed clock law, a target oscillation frequency, or a selected frozen-time eigenvalue? This version tests only whether the method and inherited diagnostics are controlled well enough to design a blind physical experiment.

3 Competing hypotheses

H_0: Apparent clock coherence may arise from projected rotation, rigid transport, beating, fit-window selection, nonconverged modes, or unqualified Euler evolution. H_1: an independently qualified PKLSA Euler vortex has an internal phase with repeatable frequency and predicts a withheld operator frequency. No physical H_0 versus H_1 discrimination is possible with the present inputs.

4 Scope and assumptions

Three-dimensional, inviscid, unforced incompressible Euler flow in a periodic domain is the model when 3D velocity fields become available. A KnotPlot centreline, a Gaussian-smoothed starting field, or a two-wavepacket detector curve is not itself a measured material-vortex phase clock. The supplied A064 data represent one geometry-only provenance group. The two covariance-gap signals are correlated, not independent physical oscillators. No SI conversion is allowed without a calibrated source length and time scale.

5 Symbols, units and dimensional consistency

Symbol	Definition	SI dimension

$\mathbf{u}$	Euler velocity	$\mathrm{m\,s^{-1}}$
$\boldsymbol{\omega}$	$\nabla \times \mathbf{u}$	$\mathrm{s^{-1}}$
ρ_f	Effective fluid density	$\mathrm{kg\,m^{-3}}$
C_{ij}	Enstrophy-weighted position covariance	$\mathrm{m^2}$
q_1, q_2	Normalized covariance eigenvalue gaps	1
ϕ_j	Unwrapped signal phase	rad
Ω_j	Phase frequency (calibrated only)	$\mathrm{rad\,s^{-1}}$

Each term of Euler acceleration has units $\mathrm{m\,s^{-2}}$; eigenvalue-gap ratios and Hilbert phases are dimensionless. Source numerical time units cannot be called physical seconds without calibration.

6 Mathematical model and observables

The upstream governing equations are the incompressible Euler system [1],

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{\nabla p}{\rho_f}, \qquad \nabla \cdot \mathbf{u} = 0. \tag{E1}$$

Vorticity is $\boldsymbol{\omega} = \nabla \times \mathbf{u}$. From squared-vorticity weights on a registered three-dimensional material region $M(t)$ one may construct the covariance C_{ij}. With ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$:

$$q_1 = \frac{\lambda_1 - \lambda_2}{\operatorname{tr} C}, \qquad q_2 = \frac{\lambda_2 - \lambda_3}{\operatorname{tr} C}. \tag{E2}$$

These variables are rotation invariant under a consistently transformed domain and region; they do not prove material-tube transport or a topological state. After centering, define a numerical analytic signal [3]

$$\phi_j(t) = \operatorname{unwrap} \arg\left[(q_j - \overline{q}_j) + i\,\mathcal{H}(q_j - \overline{q}_j)\right]. \tag{E3}$$

The measured slope (only on predeclared fitting intervals) is

$$(\widehat{a}_j, \widehat{\Omega}_j) = \arg\min_{a,b} \sum_{k \in W} \left(\phi_j(t_k) - a - b t_k\right)^2, \qquad \widehat{f_j} = \widehat{\Omega}_j/(2\pi). \tag{E4}$$

In (E4) the notation means the b component of the least-squares minimizer. Without SI-calibrated time, $\widehat{f}$ is only cycles per numerical time unit. The algorithm is a diagnostic, not a physical oscillator Hamiltonian.

Machine-readable science contract

Research question Can a source-native incompressible three-dimensional PKLSA vortex produce a reproducible intrinsic scalar phase frequency without imposing the desired frequency or time-dilation law?

Objective A068 v0.2.0 validates the diagnostic method and upstream A064 provenance. It does not identify a physical clock from geometry-only A064; physical discovery awaits a separately frozen source-qualified dynamic campaign.

H_0 H0: Any apparent periodic phase may be due to rigid-body motion, beating, windowing, translation, instrumental/spectral choices, or nonconverged Euler dynamics. Preflight controls are not physical null rejection.

H_1 H1: A certified, materially bounded, source-native Euler vortex exhibits a coherent intrinsic phase that is stable under frame reduction, time/space/core convergence and independent operator prediction. This claim is not tested physically in v0.2.0.

Registered equations

E1: Incompressible inviscid vorticity evolution

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p/\rho_f, \quad \nabla \cdot \mathbf{u} = 0 \tag{1}$$

Dimensional check: Both acceleration and pressure-gradient / density have L T^-2.

E2: Compute frame-robust phase-candidate observables

$$q_1 = (\lambda_1 - \lambda_2)/\operatorname{tr} C, \quad q_2 = (\lambda_2 - \lambda_3)/\operatorname{tr} C \tag{2}$$

Dimensional check: Eigenvalues and trace of C have L^2; ratios dimensionless.

E3: Derive measured analytic phase

$$\phi_j(t) = \arg\left[(q_j - \overline{q}_j) + i\mathcal{H}(q_j - \overline{q}_j)\right] \tag{3}$$

Dimensional check: Hilbert transform and phase are dimensionless.

E4: Recover measured frequency from linear phase trend

$$(\widehat{a}, \widehat{\Omega}) = \arg\min_{a,b} \sum_{k \in W} (\phi_k - a - bt_k)^2 \tag{4}$$

Dimensional check: Slope has T^-1; normalized source time cannot be reported as SI Hz.

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Verify user uploaded source hashes and method/simulation evidence classes exactly
2. **S2** [G2; equations E2, E3, E4]: Rerun positive sine, rigid rotation and beating null controls; cross-check archived preflight output
3. **S3** [G3; equations E1, E2]: Assess availability of source-qualified materially bounded and temporally converged PKLSA velocity evolution (none provided)
4. **S4** [G4; equations E1, E4]: Keep Euler linear/Floquet operator holdout noncertified until an actual relative-periodic background exists
5. **S5** [G5; equations E2, E4]: Require C++ FP64 and optional SYCL numerical certification only on a physical, registered dataset
6. **S6** [G7; equations E1, E4]: Require source independent replication and finite-core convergence in a later registered campaign

7 Exact algorithm and equation-to-code mapping

1. Pin framework v1.0.8; generate a fresh versioned A068 instance, preserving all earlier runs.
2. Freeze source and science contracts, report, gate graph, forbidden-term commitments; do not freeze any private reveal output in public files.

3. Verify exact SHA-256 of both preflight JSON files and A064 v0.3.0 JSON (G1).

4. Rerun the synthetic positive sinusoid and the two preregistered negative controls (rigid rotation; frequency beating) using the unmodified phase extractor (G2).

5. Audit that A064 has *not* established an Euler relative equilibrium or a source-native material PKLSA clock (G3 unresolved).

6. Do not run independent operator holdout, native C++ validation, SYCL screening or cross-provider closure without physical G3 success; record the skipped gates, not PASS.

7. Let the framework publish unique BLIND archives and its machine-generated ledger; a normal run may not reveal when unresolved physical gates remain.

8 Provenance and independence

`PREFLIGHT_RESULTS.json` and `PREFLIGHT_WITH_A064.json` are synthetic unblinded development controls. They are generated by one method family. `A064-v0.3.0_results.json` is an upstream KnotPlot-initialized Euler-Galerkin diagnostic, not an independent validated PKLSA dynamic source. The source hashes are frozen in the A068 pipeline. No source is experimental and no provider independence is claimed.

9 Blind gate definitions

Gate	Acceptance condition	Present status
G0	Framework protocol validates	Runtime conditional
G1	All three fixed hashes match	Locally reproducible
G2	Positive and both null controls reproduce	Method-only PASS candidate
G3	Materially bounded, orbit-qualified 3D PKLSA clock	UNRESOLVED
G4	Independent held-out Euler/Floquet prediction	BLOCKED by G3
G5	Native C++ FP64 physical confirmation	BLOCKED by G3/G4
G6	SYCL FP32/DD32 screening, never physical authority	BLOCKED
G7	Multiple physically independent qualified PKLSA sources	BLOCKED
G8	Material transport and space/time/core convergence	BLOCKED
G9	Valid reveal commitment after all relevant physical gates	WITHHELD

A gate cannot be marked PASS just because a prerequisite is missing, an example control is correct, or a numerical kernel exists.

10 Numerical and statistical qualification

The current synthetic positive control has input $f = 0.25$ cycles per artificial time unit and a 16-cycle record. Proposed development regression requires frequency absolute error below 10^{-5} for both signals, at least 6 observed cycles, phase-fit RMS below 0.20 rad, frequency coefficient of variation below 0.12, and relative cross-signal frequency mismatch below 0.03. These are *method thresholds*, not physical SST acceptance criteria. For the later physical experiment, $N, 2N, 4N$ meshes, time-step halving, fixed final time and independent core refinement must be separately preregistered.

11 Numerical backend authority

Python/NumPy FP64 is the method reference. Production requires separately executed strict CPU C++ FP64 certification; SYCL FP32 and DD32 are screening-only, even on an Intel Arc A770. No fake fallback may satisfy G5/G6. This instance does not yet execute native physical kernels.

12 Blindness and reveal policy

The frozen v1.0.8 runtime owns a nonced private reveal commitment, forbidden-term scans, one run ID per new blind run and non-overwriting PDFs and ZIPs. Unresolved science gates block reveal. Method controls are disclosed as synthetic; source-independent target labels and physical comparison mappings must be registered in `private/` only for a future qualified source campaign. No post-hoc physical target is read by phase extraction.

13 Statistical design and multiplicity

Synthetic controls are deterministic regression checks, not a population of independent source replicates. Consequently, no physical p-value, false-discovery-rate, significance claim, or cross-topology confidence interval is reported. A later discovery/holdout study must choose independent source units, sample sizes, fixed windows and multiplicity corrections before examining outcomes.

14 Results and output contract

Source and method checks are rerun by the scientific overlay and recorded in the framework gate ledger / backend manifest. For each BASIC/FULL/CERTIFY run the v1.0.8 framework archives prior work and publishes a new run-identified BLIND ZIP and PDF when LaTeX is available.

Automatic run provenance

Field	Recorded value
Run ID	20261017T204511996339Z-c9842f36
Run phase	BLIND
Protocol bundle SHA-256	276d704d63d8c7e1add1d075b959102cddf08047600006e2bb07d74068203ac0
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	c50b6dd3b5b3140ceff8b06f5480ecd4b88447ce60224890a5e50c87544e8c4b

Gate ledger

Gate	Status	Question
G0	PASS	Has framework v1.0.8 verified the exactly pinned, frozen science and b
G1	PASS	Are the three preflight/A064 method-audit source files byte-identical a
G2	PASS	Does Python FP64 reproduce the synthetic positive and negative phas
G3	UNRESOLVED	Is there a source-native material-bounded, converged, relative-periodic
G4	NOT_RUN_PREREQUISITE	Does an independent preregistered Euler/Floquet operator predict a p
G5	NOT_RUN_PREREQUISITE	Is physical numerical extraction independently reproduced by native C
G6	NOT_RUN_PREREQUISITE	Does SYCL/Arc parity screen only (FP32/DD32), without being prom
G7	NOT_RUN_PREREQUISITE	Does the physical result replicate across truly independent qualified Pl

G8	NOT_RUN_PREREQUISITE	Are material transport, finite-core, time/space convergence and physic
G9	DEFERRED	After successful gate dependency chain, does revealed SST clock comp

Backend manifest


```
{
  "actual_backend": "python-numpy-fp64",
  "authority": "METHOD_DEVELOPMENT_ONLY",
  "no_independent_pklsa_provider_claim": true,
  "no_native_cpp_or_sycl_claim": true,
  "optional_3d_field": {
    "reason": "No time-resolved 3D velocity field supplied; A064 is not that dataset.",
    "status": "NOT_PROVIDED"
  },
  "physical_clock_status": "UNRESOLVED",
  "precision": "float64",
  "preflight_result": {
    "a064_audit": {
      "G0_real_PKLSA_clock_data": "UNRESOLVED_GEOMETRY_ONLY",
      "G1_periodic_relative_equilibrium": "UNRESOLVED",
      "G2_two_independent_intrinsic_3D_shape_signals": "NOT_PROVIDED",
      "G3_phase_frequency": "NOT_TESTABLE",
      "G4_operator_prediction_holdout": "NOT_TESTABLE",
      "G5_mesh_timestep_core_convergence": "UNRESOLVED",
      "G6_independent_provider": "UNRESOLVED",
      "overall": "NO_PHYSICAL_CLOCK_INFERENCE_POSSIBLE",
      "prior_conclusion": "NUMERICAL_OPERATOR_EIGENPAIRS_VERIFIED; G6 UNRESOLVED; G7 WAVE_PROP
      "reported_physical_gates": {
        "G6a_steady_relative_equilibrium": "UNRESOLVED (background is nonstationary)",
        "G6b_asymptotic_spectral_convergence": "UNRESOLVED (n=8/12/16 extremal branches differ
        "G6c_Kelvin_eigenbundle_connection": "UNRESOLVED (no unique Kelvin eigenspace transpor
        "G7a_both_counterpropagating": "SCREEN_FAIL (tested launch does not establish both bra
        "G7b_recombination": "SCREEN_FAIL (no synchronized verified arrival)",
        "G7c_topology_specific_interference": "UNRESOLVED (no causal topology isolation)",
        "G8_PKLSA_provider_independence": "UNRESOLVED (single geometry-only provider)",
        "G9_official_framework_blind_native": "UNRESOLVED (v1.0.7 referenced but not invoked)"
      },
      "sha256": "3c1469cfddeafbc69c368a779072144119769d5ee9276a92064c803fbc6af3c1",
      "source": "A064-v0.3.0_results.json"
    },
    "checks": {
      "A064_identity_matches": true,
      "A064_no_physical_clock_claim": true,
      "A064_periodic_background_not_certified": true,
      "beating_rejected": true,
      "original_demo_reproducible": true,
      "original_is_unblinded_method": true,
      "positive_numeric_clock_recovered": true,
      "rigid_rotating_shape_rejected": true
    },
    "interpretation": "SYNTHETIC_METHOD_VALIDATION_ONLY",
    "pass_method_control": true,
    "synthetic_frequency": 0.2499999819538135,
    "synthetic_omega": 1.5707962134073628
```

```
  },
  "project": "A068_PKLSA_PhaseClock_v0.2.0",
  "requested_framework": "SST_Falsifier_Framework_v1.0.8",
  "schema": "SST-BACKEND-MANIFEST-2",
  "source_hash_status": {
    "all_valid": true,
    "files": {
      "A064-v0.3.0_results.json": {
        "actual_sha256": "3c1469cfddeafbc69c368a779072144119769d5ee9276a92064c803fbc6af3c1",
        "exists": true,
        "expected_sha256": "3c1469cfddeafbc69c368a779072144119769d5ee9276a92064c803fbc6af3c1",
        "sha256_matches": true
      },
      "PREFLIGHT_RESULTS.json": {
        "actual_sha256": "eb2c099968b5ab2b33abb9ce855b0a4c77f58cd0f0137ccabdaf917146574730",
        "exists": true,
        "expected_sha256": "eb2c099968b5ab2b33abb9ce855b0a4c77f58cd0f0137ccabdaf917146574730",
        "sha256_matches": true
      },
      "PREFLIGHT_WITH_A064.json": {
        "actual_sha256": "7f6f865af5e6c459462bcc88780415a1d7257e96558cc47cae9ac6bd723f6902",
        "exists": true,
        "expected_sha256": "7f6f865af5e6c459462bcc88780415a1d7257e96558cc47cae9ac6bd723f6902",
        "sha256_matches": true
      }
    }
  }
}
```


Source manifest


```
{
  "manifest_sha256": "924fe0f779c81fdd382d3222f2c302050c5a23f84fdfc3ff117927fee025d83a",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A068",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic_preflight_single_generator",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A068_PKLSA_P
      "provenance_family": "synthetic_preflight_single_generator",
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "method_control",
      "sha256": "eb2c099968b5ab2b33abb9ce855b0a4c77f58cd0f0137ccabdaf917146574730",
      "source_id": "A068_CONTROL_A",
      "version": "v0.1.0"
    },
    {
      "catalog_id": "A068",
      "evidence_class": "synthetic_control",
```

```
      "exists": true,
      "independence_family": "synthetic_preflight_single_generator",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A068_PKLSA_P
      "provenance_family": "synthetic_preflight_single_generator",
      "registry_hint": {
        "catalog_id": "A068",
        "hits": []
      },
      "role": "prior_diagnostic",
      "sha256": "7f6f865af5e6c459462bcc88780415a1d7257e96558cc47cae9ac6bd723f6902",
      "source_id": "A068_CONTROL_B",
      "version": "v0.1.0"
    },
    {
      "catalog_id": "A064",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "knotplot_catalogue_and_procedural_native",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A068_PKLSA_P
      "provenance_family": "knotplot_catalogue_and_procedural_native",
      "registry_hint": {
        "catalog_id": "A064",
        "hits": []
      },
      "role": "upstream_geometry_initiated_simulation_audit",
      "sha256": "3c1469cfddeafbc69c368a779072144119769d5ee9276a92064c803fbc6af3c1",
      "source_id": "A064_UPSTREAM",
      "version": "v0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Interpretation and falsification limits

A synthetic control PASS means the diagnostic is working on that control; it does not validate vortex clocks. A064's instantaneous eigenpairs are not a time-harmonic Kelvin basis, and no observed velocity field has established a material knotted vortex oscillator. Failures of one candidate mode or packet launch cannot falsify every vortex-frequency mechanism. A consistent time-dependent, materially supported and converged field would permit the next physical campaign, not retroactive promotion of these data.

16 Reproducibility and references

Install using `install_A068_v0.2.0.py` with a real local Workbench containing v1.0.8. The installer invokes the *official generator*, writes a matching relative framework pin, replaces only the new instance files, and refuses an existing target version. Test via `pytest -q tests`; then `run_all.cmd FREEZE`, `run_all.cmd BASIC`, and `run_all.cmd REVEAL_IF_ALLOWED`. The archived SHA-256 inputs and this report are included in the distribution.

References

[1] P. G. Saffman (1992), *Vortex Dynamics*, Cambridge University Press, doi:10.1017/CBO9780511624063.

[2] H. K. Moffatt (1969), "The degree of knottedness of tangled vortex lines," *Journal of Fluid Mechanics* **35**, 117–129, doi:10.1017/S0022112069000991.

[3] D. Gabor (1946), "Theory of communication," *Journal of the Institution of Electrical Engineers*, Part III **93**, 429–457, doi:10.1049/ji-3-2.1946.0074.